Dr. Alexander Nettekoven

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Role: Postdoc @ UTexas
US Work: No Restrictions
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Education

2019 – 2022 **University of Texas at Austin**, Austin, TX
Ph.D. in Mechanical Engineering (Specialization: Robotics)

2016 – 2018 **University of Texas at Austin**, Austin, TX
M.Sc. in Mechanical Engineering

2011 – 2016 **RWTH Aachen University**, Aachen, Germany
B.Sc. in Mechanical Engineering

Professional Experience

06/2025 – Present **Center for Autonomy**, UT Austin – **Postdoctoral Fellow**
Systematic evaluation of state-of-the-art robot foundation models (VLA models, World Action Models) in simulation and on a low-cost bimanual manipulator; benchmarking policy performance and identifying failure sources.

09/2022 – 06/2025 **Multi AI, Inc.**, Austin, TX / San Francisco, CA – **Co-Founder & CEO**
Grew company from PhD-stage prototype to Pentagon-contracted product; built multi-agent AI platform (decentralized MCTS + generative AI) achieving 38% improvement in disrupted logistics scenarios.

11/2021 – 09/2022 **Droneconia LLC** (later Multi AI, Inc.), Austin, TX – **Co-Founder & Chief Technology Officer**
Designed initial multi-agent AI algorithms for autonomous UAV/UGV mission planning; secured and delivered Air Force SBIR Phase I contract from AFWERX Agility Prime.

01/2019 – 12/2022 **Autonomous Systems Group**, UT Austin – **NSF-Supported Researcher**
Research on autonomous systems and robotics in NSF-funded collaboration with Caltech, Berkeley, Northeastern, and RPI.

08/2016 – 12/2018 **Advanced Manufacturing Center**, UT Austin – **Research Assistant**
Designed and commissioned hardware testbeds for advanced manufacturing processes; developed data-driven control and anomaly detection pipelines for automated process monitoring.

01/2016 – 04/2016 **Mercedes-Benz**, Stuttgart, Germany – **Robot Operations Intern**
ABB robotic assembly optimization for the OM-651 diesel engine at Daimler headquarters.

11/2015 – 12/2015 **Ford Germany**, Cologne, Germany – **Maintenance Intern**
Maintenance and fault diagnosis of ABB industrial robots in aluminum die casting production.

04/2015 – 11/2015 **Fraunhofer Institute of Laser Technology**, Aachen, Germany – **Research Assistant**
Fabricated iron-aluminide alloy specimens via Laser Metal Deposition (LMD), achieving densities >99.5%; programmed end-effector trajectories and managed laser safety protocols.

10/2014 – 02/2015 **University of Michigan**, Ann Arbor, MI – **Undergraduate Research Scholar**
Signal detection for Electrohydrodynamic Jet (e-jet) printing in Prof. Kira Barton's lab.

Achievements & Recognitions

2024 **Pentagon Wargaming Development Partnership**. Six-figure SBIR Phase III by Chief of Wargaming, OUSD A-S (Pentagon), and Air Force Autonomy Prime.

2024 **Invited Expert to Texas AI Caucus**. Advised Texas State Representative Giovanni Capriglione on AI policy and commercial regulations.

2024 **Seven-Figure TACFI Selection by US Air Force**. Awarded for highly-selective Funding Increase program by Air Force Ventures for decision-support AI.

2023 **Major Phase III Award by US Air Force Autonomy Prime**. Six-figure contract for multi-agent coordination AI.

2023 **Capital Factory Accelerator**. Selected for Texas's largest and most active startup accelerator.

2022 **Launch Texas Awardee**. Launch Texas awardee of newly-founded space entrepreneurship program at UT Austin.

2022 **Major Phase II Award by US Air Force Agility Prime**. Secured first six-figure contract for multi-agent coordination AI to accelerate rapid scaling in DoD.

2022 **Selection for Elite Startup Accelerators**. MassChallenge finalist out of 400 startups and selection to exclusive Air Force Technology Acceleration Program.

2022 **Texas Venture Labs Commercialization Award Winner**. Five-figure prize for highest commercialization success among Texas student startups.

2022 **Bridge-to-Phase II Award Winner**. UT Discovery grant for accelerating national security decision-support AI.

2022 **Major Air Force Programs and Prime Support**. Formal endorsements from AFWERX, AFRL ACT3, RQQ, RISB, and Textron for startup's solution offering.

2021 **GAIN Mechanical Engineering Department Award**. Outstanding research contributions award at University of Texas at Austin.

2018-2020 **National Science Foundation (NSF)-Supported Researcher**. Selected researcher on multi-year project with CalTech, Berkeley, Northeastern, and RPI.

2017 **Sandia National Laboratories Grant**. Proposed and won five-figure funding for innovative laser pre-scan process enhancing 3D printing quality.

2015 **Undergraduate Research Opportunity Award**. Dual award by the University of Michigan and RWTH Aachen University for undergraduate research.

Publications

2024 **Alexander Nettekoven**, Matthew Carlin, and Ufuk Topcu. "Agile Combat Employment System (ACES) Model Modifications." *Technical Report to US Air Force AFWERX SBIR Phase III.*

2023 **Alexander Nettekoven**, Matthew Carlin, and Ufuk Topcu. "Large-Scale Fleet Coordination for Contested Logistics." *Technical Report to US Air Force AFWERX SBIR Phase II.*

2022 **Alexander Nettekoven**. "On the Feasibility of 3D Printing with Multicopters." *Ph.D. Dissertation at the University of Texas at Austin.*

2022 **Alexander Nettekoven**, Scott Fish, Joseph Beaman, and Ufuk Topcu. "Towards online monitoring and data-driven control: a study of segmentation algorithms for infrared images of the powder bed." *2022 International Solid Freeform Fabrication Symposium.*

2021 **Alexander Nettekoven**, Nicholas Franken, and Ufuk Topcu. "Geometrical Analysis of Simple Contours Deposited by a 3D Printing Hexacopter." *2021 International Solid Freeform Fabrication Symposium.*

2021 **Alexander Nettekoven** and Ufuk Topcu. "A 3D Printing Hexacopter: Design and Documentation." *2021 International Conference on Unmanned Aircraft Systems (ICUAS).*

2021 Zizhao Wang, Xuesu Xiao, **Alexander J. Nettekoven**, Kadhiravan Umasankar, Anika Singh, Sriram Bommakanti, Ufuk Topcu, and Peter Stone. "From agile ground to aerial navigation: Learning from learned hallucination." *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).*

2019 Yingshui Tan, Baihong Jin, **Alexander Nettekoven**, Yuxin Chen, Yisong Yue, Ufuk Topcu, and Alberto Sangiovanni-Vincentelli. "An encoder-decoder based approach for anomaly detection with application in additive manufacturing." *2019 18th IEEE International Conference on Machine Learning and Applications (ICMLA).*

2019 Zhe Xu, **Alexander J. Nettekoven**, A. Agung Julius, and Ufuk Topcu. "Graph temporal logic inference for classification and identification." *2019 IEEE 58th Conference on Decision and Control (CDC).*

2018 **Alexander Nettekoven**. "Predictive iterative learning control with data-driven model for near-optimal laser power in selective laser sintering." *2018 Master's Thesis.*

2018 **Alexander Nettekoven**, Scott Fish, Joseph Beaman, and Ufuk Topcu. "Predictive Iterative Learning Control with Data-Driven Model for Optimal Laser Power in Selective Laser Sintering." *2018 International Solid Freeform Fabrication Symposium.*

Services

2021 – present **Peer Reviewer for Robotics and Engineering Conferences:**

- International Conference on Robotics and Automation (ICRA)
- International Conference on Intelligent Robots and Systems (IROS)
- IEEE Robotics and Automation Letters (RA-L)
- American Control Conference (ACC)
- North American Manufacturing Research Conference (NAMRC)
- Solid Freeform Fabrication Symposium (SFF)
- Conference on Automation Science and Engineering (CASE)

2020 – present **Guest Lecturer for Undergraduate Feedback Control**. Presentations of real-world control applications to engineering students at UT Austin.

2025 **STEM Outreach, Del Valle ISD**. Taught students to build and program a self-driving RC car.

2021 **Texas Robotics Motion Capture Space**. Designed, directed, and commissioned high-resolution motion capture space for UT Austin robotics program.

2021 **Public Safety Drone Collaboration Workshop**. Organized Drone workshop for Austin PD, Texas DPS, Travis County Sheriff's Office, and Austin Fire.

2020 – 2022 **Research Mentor**. Supervised 7 undergraduate research assistants: Sriram Bommakanti, Kadhir Umasankar, Nicholas Franken, Harrison Jin, Meenakshi Swaminathan, Steven Diep, and Mario Gonzalez.

2020 **Research Instructor**. Led full-time summer research internships for 17 undergraduate students in the Autonomous Systems Group.

Press Coverage

2023 Jeremey R. Dunn. "Tech Connect partner uses AI to assist in in-theater mission planning." *Defense Visual Information Distribution Service* (08/23/2023).

2023 Deborah Yao. "Hybrid AI Reduces Generative AI Hallucinations, Applied Intelligence Live! Austin 2023." *AI Business* (09/21/2023).

2022 Nat Levy. "Ready for Launch." *Texas Engineer* (2022-2023 issue).

Invited Talks and Presentations

12/2024 US Air Force Wargaming Offices and OUSD A-S Chief of Wargaming Workshop.

03/2024 HAF/a5, SAF/SA, HAF/A4, RGPA, AMC A5-8, AMC A9 Wargaming Capabiltiy Review.

09/2023 Capital Factory Accelerator First Look Kickoff.

08/2023 National Security Innovation Council (NSIC).

05/2023 Texas Innovation Center Board Meeting and Luncheon.

11/2022 Launch Texas Award and Kickoff at the University of Texas at Austin.

09/2022 Blue Tech Collider Multi-Drone Show Case.

08/2022 Solid Freeform Fabrication Conference 2022.

07/2022 MassChallenge Early-Stage Startup Accelerator Kickoff.

02/2022 Hanscom and Incirlik US Air Force Base Capability Review.

08/2021 Solid Freeform Fabrication Conference 2021.

06/2021 International Conference on Unmanned Aircraft Systems.

04/2021 Texas Robotics Research Seminar.

02/2021 Graduate and Industry Networking event, University of Texas at Austin.

10/2020 Walker Department of Mechanical Engineering, University of Texas at Austin.

03/2019 NSF Cyberphysical System Review with Caltech, Berkeley, Northeastern, RPI.

08/2018 Solid Freeform Fabrication Conference 2018.

02/2017 America Makes Project Review with US Air Force AFRL.

06/2016 International Center for Sustainable Textiles at RWTH Aachen University.

02/2015 Barton Research Group at the University of Michigan.